Computer Networking — Study Notes

A sample reference document for testing RAG pipelines. Covers core transport-layer, addressing, and security concepts you'd expect on an intro networking exam.

1. Transport Layer Protocols

1.1 TCP (Transmission Control Protocol)

TCP is a connection-oriented protocol. Before data is sent, the client and server perform a three-way handshake (SYN, SYN-ACK, ACK) to establish a connection. TCP guarantees reliable, ordered delivery of data: lost packets are retransmitted, and packets are reassembled in the correct order at the destination. This reliability comes at the cost of higher latency and overhead. TCP is used where data integrity matters more than speed — examples include web browsing (HTTP/HTTPS), email (SMTP), and file transfer (FTP).

1.2 UDP (User Datagram Protocol)

UDP is connectionless — there is no handshake, and no guarantee that packets arrive, arrive in order, or arrive only once. This makes UDP much faster and lower-overhead than TCP. UDP is preferred for real-time applications where speed matters more than perfect accuracy, such as video streaming, online gaming, and VoIP calls. A dropped UDP packet is simply skipped rather than retransmitted.

1.3 Key Difference

The core tradeoff: TCP trades speed for reliability, UDP trades reliability for speed. TCP header size is 20 bytes minimum; UDP header is only 8 bytes, which is part of why it's faster.

2. IP Addressing

2.1 IPv4 vs IPv6

IPv4 addresses are 32-bit, written as four decimal numbers separated by dots (e.g. 192.168.1.1), giving about 4.3 billion possible addresses. IPv6 addresses are 128-bit, written in hexadecimal groups separated by colons (e.g. 2001:0db8::1), created to solve IPv4 address exhaustion.

2.2 Public vs Private IP Addresses

Private IP ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) are reserved for use inside local networks and are not routable on the public internet. A router uses NAT (Network Address Translation) to translate private addresses to a single public IP when devices communicate externally.

2.3 Subnetting

Subnetting divides a network into smaller sub-networks using a subnet mask (e.g. /24 = 255.255.255.0). This improves routing efficiency and network security by isolating traffic into smaller segments.

3. Network Security Basics

3.1 Firewalls

A firewall monitors and filters incoming and outgoing traffic based on security rules. It can operate at the packet level (filtering by IP/port) or at the application level (inspecting actual content).

3.2 Encryption in Transit

HTTPS uses TLS (Transport Layer Security) to encrypt data between client and server, preventing eavesdropping and man-in-the-middle attacks. TLS also verifies server identity via digital certificates issued by a trusted Certificate Authority (CA).

3.3 DNS and DNS Spoofing

DNS (Domain Name System) translates human-readable domain names into IP addresses. DNS spoofing (cache poisoning) is an attack where false DNS records redirect users to malicious sites. DNSSEC adds cryptographic signatures to DNS responses to prevent this.